

FORT LIARD, NT, Canada

WMO: 714970

Lat: 60.236N

Lon: 123.473W

Elev: 700

StdP: 14.33

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.5	-29.5	-43.8	0.4	-35.2	-37.6	0.7	-29.5	21.1	6.7	18.9	4.7	6.1	220	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	83.5	63.0	80.0	61.6	76.8	60.1	66.3	78.0	64.3	75.4	62.4	72.9	5.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.1	85.4	70.5	60.1	79.6	68.4	58.2	74.2	66.4	31.4	77.8	29.8	75.4	28.5	73.2	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.4	16.3	14.3	DB	-40.4	89.6	6.3	3.0	-45.0	91.8	-48.6	93.6	-52.2	95.3	-56.7	97.5	
			WB	-40.1	71.2	6.3	2.9	-44.6	73.3	-48.3	75.0	-51.9	76.6	-56.4	78.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.1	-6.1	3.7	14.2	35.6	50.5	59.6	63.6	60.1	49.5	33.5	9.3	-2.4	(6)
	DBStd	27.22	15.56	13.54	14.60	10.03	8.63	5.95	5.29	6.28	7.46	8.77	12.80	14.08	(7)
	HDD50	8152	1740	1297	1110	435	98	5	1	6	99	514	1222	1625	(8)
	HDD65	12479	2205	1717	1575	883	452	178	89	175	467	976	1672	2090	(9)
	CDD50	1235	0	0	0	2	113	294	422	319	82	3	0	0	(10)
	CDD65	87	0	0	0	0	3	16	45	22	1	0	0	0	(11)
	CDH74	1337	0	0	0	0	124	338	552	296	25	2	0	0	(12)
	CDH80	297	0	0	0	0	22	74	134	63	4	0	0	0	(13)
	WSAvg	6.2	6.7	6.7	7.4	6.9	6.4	5.5	5.0	5.2	5.5	5.8	6.2	6.7	(14)
	PrecAvg	17.4	1.0	0.8	0.6	0.7	1.6	2.7	3.1	2.1	2.0	1.2	1.3	1.0	(15)
(16) Precipitation	PrecMax	24.5	3.1	3.0	1.7	2.1	3.3	5.6	9.4	4.3	4.2	3.2	3.5	2.8	(16)
	PrecMin	13.6	0.2	0.2	0.0	0.0	0.2	0.8	0.4	0.8	0.3	0.3	0.1	0.3	(17)
	PrecStd	3.0	0.9	0.6	0.4	0.5	0.8	1.5	2.0	1.0	1.2	0.8	0.8	0.8	(18)
	PrecStd	3.0	0.9	0.6	0.4	0.5	0.8	1.5	2.0	1.0	1.2	0.8	0.8	0.8	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	46.6	55.4	67.3	82.9	86.5	89.1	86.3	76.8	63.5	46.5	50.2	(19)
		MCWB	36.6	36.2	41.4	48.6	57.6	61.9	66.1	67.0	59.6	49.4	36.9	40.6	(20)
	2%	DB	33.6	37.6	48.2	61.8	77.4	81.9	83.6	80.9	70.9	56.2	39.0	34.6	(21)
		MCWB	29.2	30.6	37.8	45.7	55.4	60.9	64.2	64.1	55.3	44.4	32.7	29.8	(22)
	5%	DB	21.7	28.7	43.2	57.4	72.8	77.7	80.0	77.0	67.3	51.3	31.2	24.6	(23)
		MCWB	19.8	25.1	34.5	42.8	53.5	59.1	62.9	61.6	53.5	42.3	28.2	22.6	(24)
	10%	DB	13.1	22.1	36.2	53.3	68.1	74.2	76.5	73.3	63.0	46.0	25.8	16.3	(25)
		MCWB	11.9	19.9	30.6	40.8	51.1	57.6	61.6	60.0	51.6	39.8	24.0	15.2	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.2	36.6	42.5	49.4	60.0	66.1	69.5	69.7	61.2	50.7	36.8	41.3	(27)
		MCDB	43.5	45.9	54.8	65.2	76.1	79.4	82.5	82.3	72.8	60.8	45.4	49.5	(28)
	2%	WB	29.8	30.9	38.1	46.2	57.2	63.5	66.7	66.2	57.6	45.8	32.4	30.5	(29)
		MCDB	34.4	37.2	47.6	60.6	73.5	75.7	78.7	76.3	68.3	53.8	38.3	34.7	(30)
	5%	WB	20.2	25.3	34.7	43.5	54.8	61.7	64.9	63.6	54.9	43.0	27.8	22.1	(31)
		MCDB	22.0	28.8	42.7	55.8	69.9	73.3	76.2	74.0	63.9	50.7	30.7	24.5	(32)
	10%	WB	12.1	20.0	30.7	41.1	52.5	59.7	63.1	61.3	52.8	40.1	23.7	14.7	(33)
		MCDB	13.2	22.2	36.4	52.5	66.4	70.5	73.5	70.7	61.0	45.7	25.4	15.8	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	13.3	18.8	23.2	22.8	24.4	22.7	21.4	21.4	20.0	13.5	12.6	12.9	(35)
		MCDBR	24.5	27.9	31.6	31.1	34.0	30.7	28.8	28.6	27.5	24.0	19.8	24.3	(36)
		MCWBR	20.5	21.9	21.4	17.1	15.3	13.1	12.8	13.7	14.3	14.0	15.2	20.5	(37)
	5% WB	MCDBR	24.6	27.4	30.8	29.7	31.1	26.2	26.1	26.5	25.8	22.5	19.0	24.3	(38)
		MCWBR	20.3	21.8	20.9	16.8	14.6	12.2	12.7	13.8	14.5	13.4	14.4	19.9	(39)
(40) Clear-Sky Solar Irradiance	taub		0.212	0.245	0.272	0.335	0.349	0.363	0.370	0.361	0.315	0.287	0.223	0.177	(40)
	taud		2.171	2.306	2.272	2.228	2.416	2.430	2.422	2.462	2.581	2.513	2.201	2.042	(41)
	Ebn at Noon		190	250	275	273	279	276	270	264	262	233	180	154	(42)
	Edh at Noon		12	20	28	36	33	33	32	29	22	16	11	9	(43)
(44) All-Sky Solar Radiation	RadAvg		134	415	953	1496	1751	1808	1723	1391	894	419	155	71	(44)
	RadStd		10	31	63	67	117	136	135	112	91	40	19	5	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.21	+4.30	+6.19	N/A	N/A	N/A	-304	-416	N/A	+27
Regional (5 neighbors)		N/A	+3.44	+4.50	N/A	N/A	N/A	N/A	N/A	+117	N/A

Nomenclature: See separate page